An Identity Relating Triangular and Pentagonal Numbers

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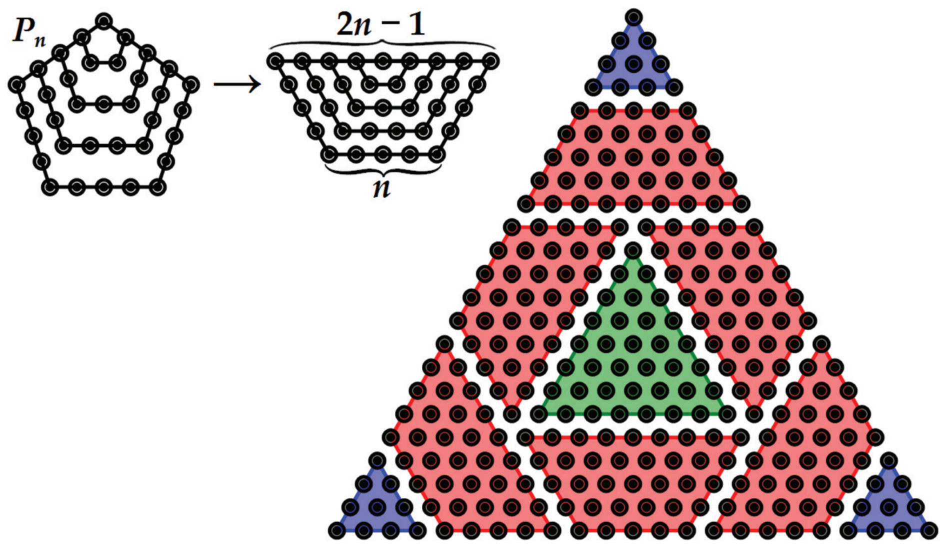
Theorem 1. *Let P_n and T_n represent the nth pentagonal and the nth triangular number, respectively, for $n \in \mathbb{N}$. Then we have the following identity:*

$$T_{5n-2} = T_{2n-2} + 6P_n + 3T_{n-1}.$$

We illustrate the proof for $n = 5$.



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Summary. We give a visual proof for an identity relating triangular and pentagonal numbers.

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